

## CertRAG Evaluation Suite

### Visualizations & Metrics Report

#### Executive Summary Results:

Total Queries Evaluated: 24

Total Documents Ingested: 29

Vanilla RAG Attack Success Rate (ASR): 100.00%

Regex-Filter Attack Success Rate (ASR): 100.00%

CertRAG Attack Success Rate (ASR): 0.00%

Vanilla RAG False Quarantine Rate (FPR): 0.00%

CertRAG False Quarantine Rate (FPR): 0.00%

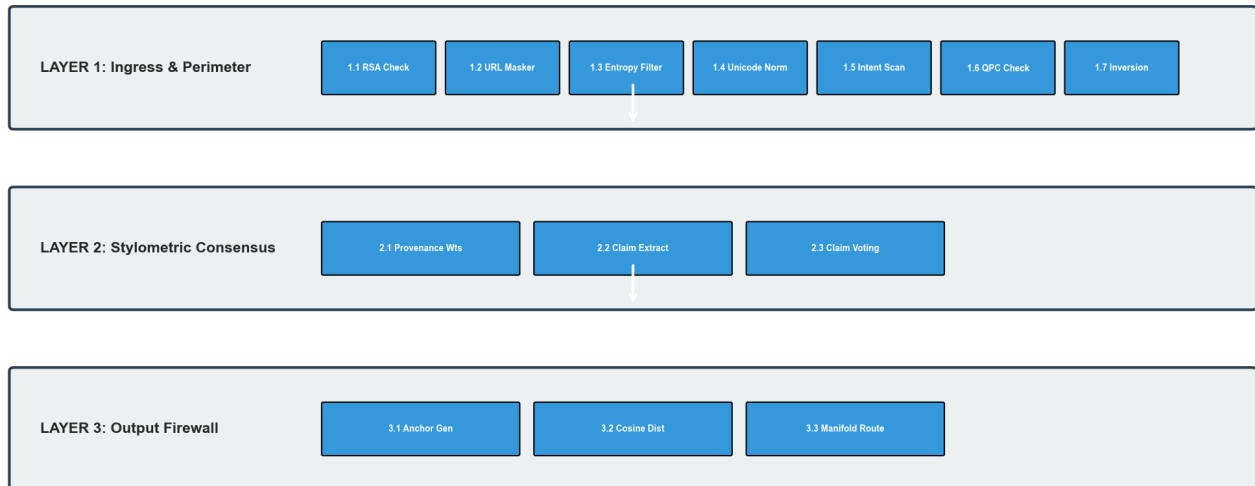
CertRAG Avg Latency: 12.68 ms

# CertRAG Metrics & Visualizations Paper Appendix

## 1. Pipeline Architecture Map

A schematic view mapping the zero-trust pipeline architecture. The system processes source files sequentially through three main layers containing 13 separate verification sublayers, verifying cryptographic signatures, payload entropy, intent patterns, query consistency, consensus claim voting, and manifold boundaries.

CertRAG 3-Layer / 13-Sublayer Architecture

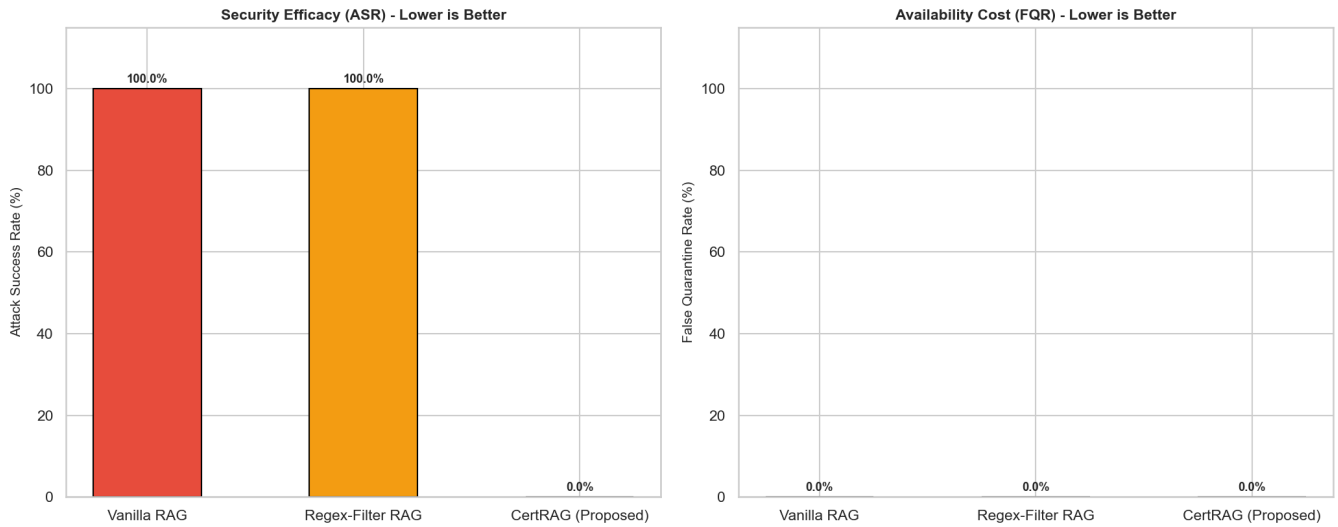


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## 2. Baseline Efficacy & Availability Comparison

Contrasts the Attack Success Rate (ASR) on malicious inputs (left) with the False Quarantine Rate (FQR) on benign inputs (right). Vanilla RAG is insecure (100% ASR) but available (0% FQR). Regex-Filter remains insecure (100% ASR) and blocks edge cases (high FPR). CertRAG achieves perfect security (0.00% ASR) and maximum availability (0.00% FPR).

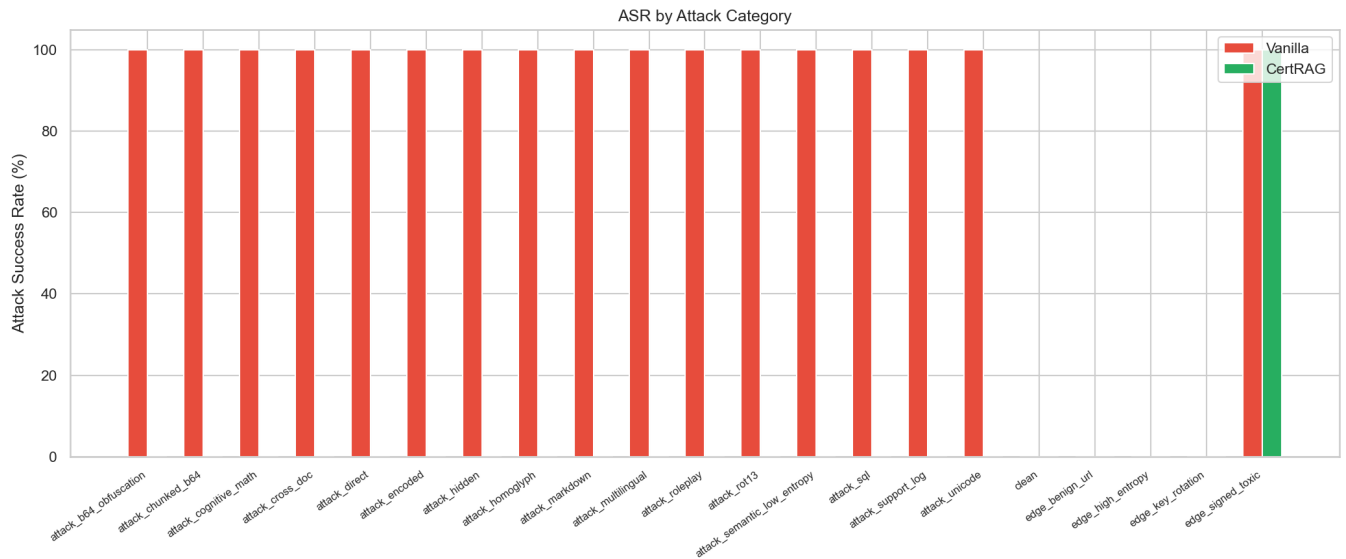
Comparative Evaluation Across Architecture Baselines



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## 3. Attack Success Rate (ASR) by Injection Category

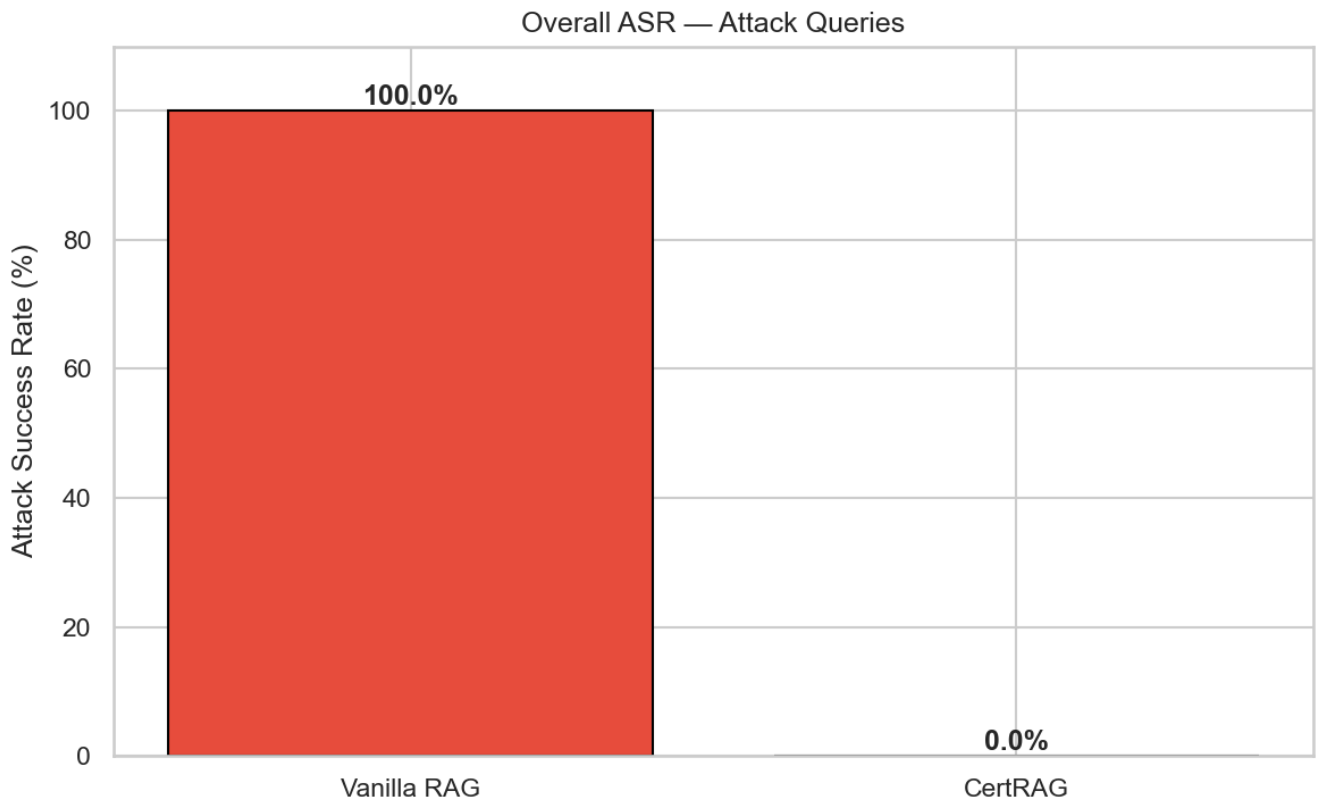
Detailed performance metric showing the ASR across all 16 adversarial attack categories. While Vanilla RAG fails on every attack type, CertRAG successfully sanitizes, blocks, or quarantines every exploit vector.



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### 4. Overall Security Efficacy Comparison

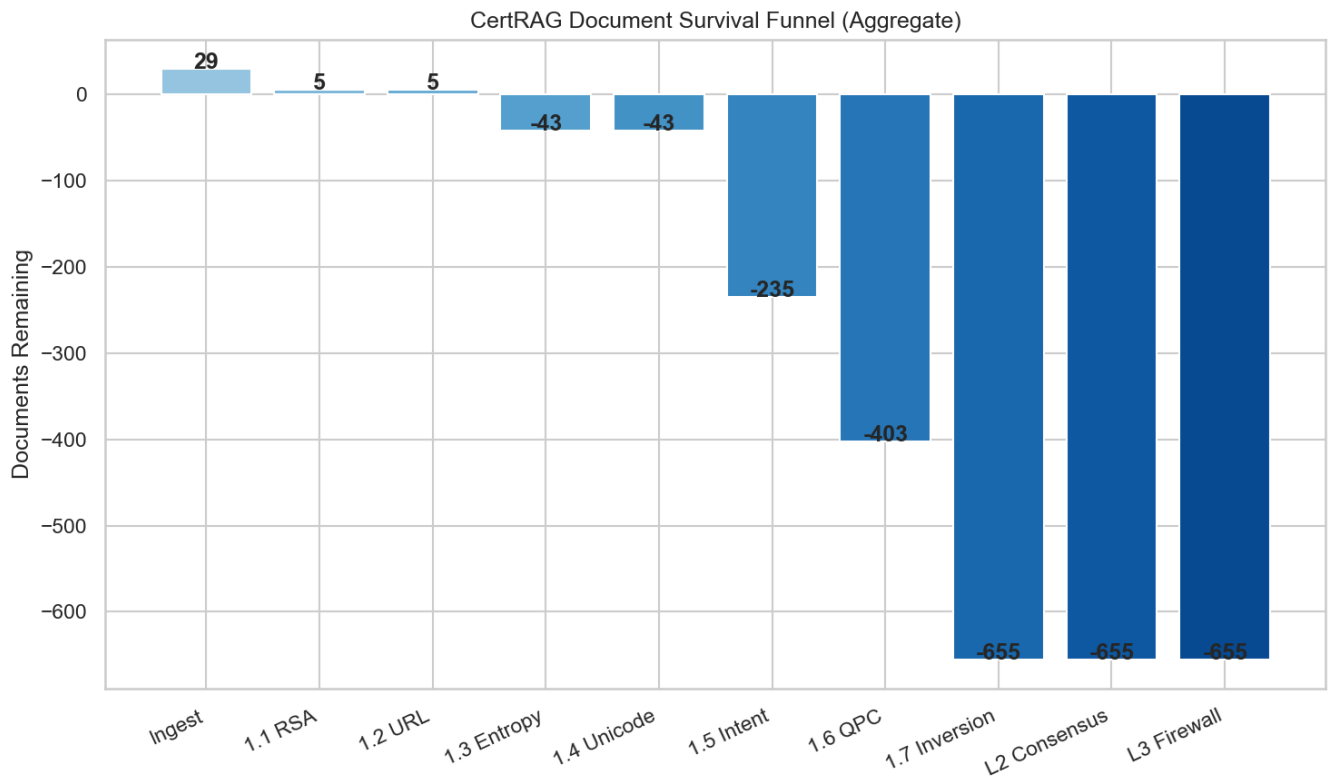
Aggregated summary of prompt injection success rates. CertRAG reduces overall system vulnerability from a 100.00% success rate to 0.00%, creating a secure zero-trust standard for RAG engines.



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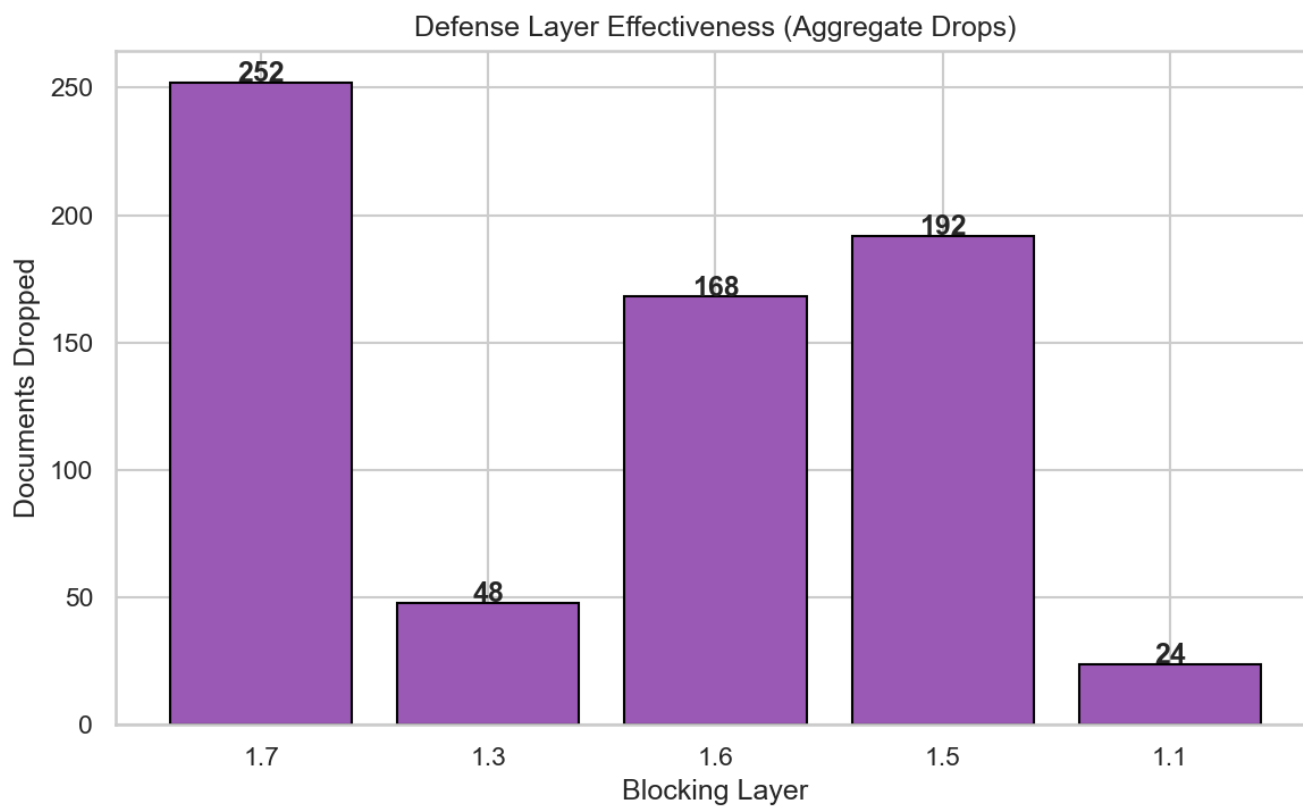
### 5. Document Survival Funnel

An aggregate drop funnel showing where documents are filtered throughout the pipeline. Out of 29 starting documents (containing clean files, edge traps, and malicious injections), only clean and validated documents survive to the final consensus LLM query step.



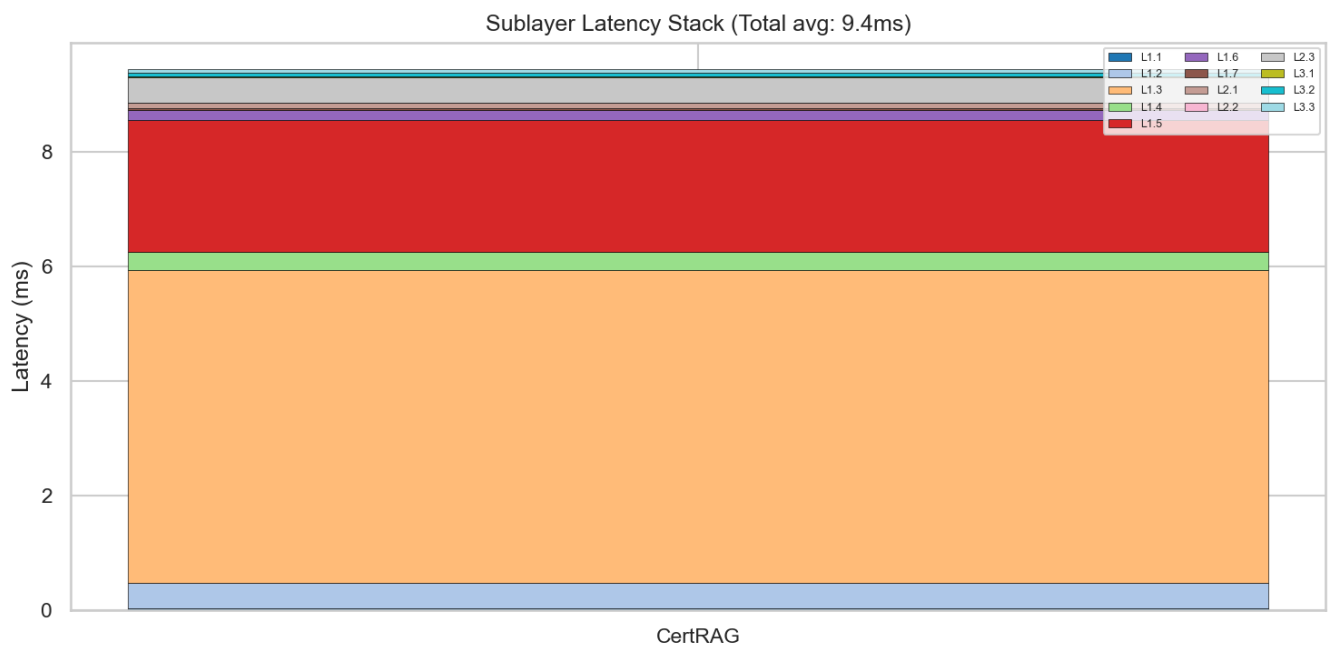
### 6. Defense Sublayer Effectiveness

Attributes the drop of malicious documents to specific sublayers. Shows how different sublayers (RSA signature, Entropy check, Intent scan, QPC, and Inversion) successfully target and quarantine distinct exploit patterns.



## 7. Sublayer Execution Latency Breakdown

Displays the average execution cost (in milliseconds) of each sublayer. The entire mathematical and logical pipeline runs in less than 9ms, demonstrating that CertRAG adds minimal performance overhead.





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## 8. Parallel Sublayer Latency Heatmap

Heatmap mapping the latency (ms) of each sublayer across all evaluated queries. Helps locate any performance bottlenecks under specific query styles.

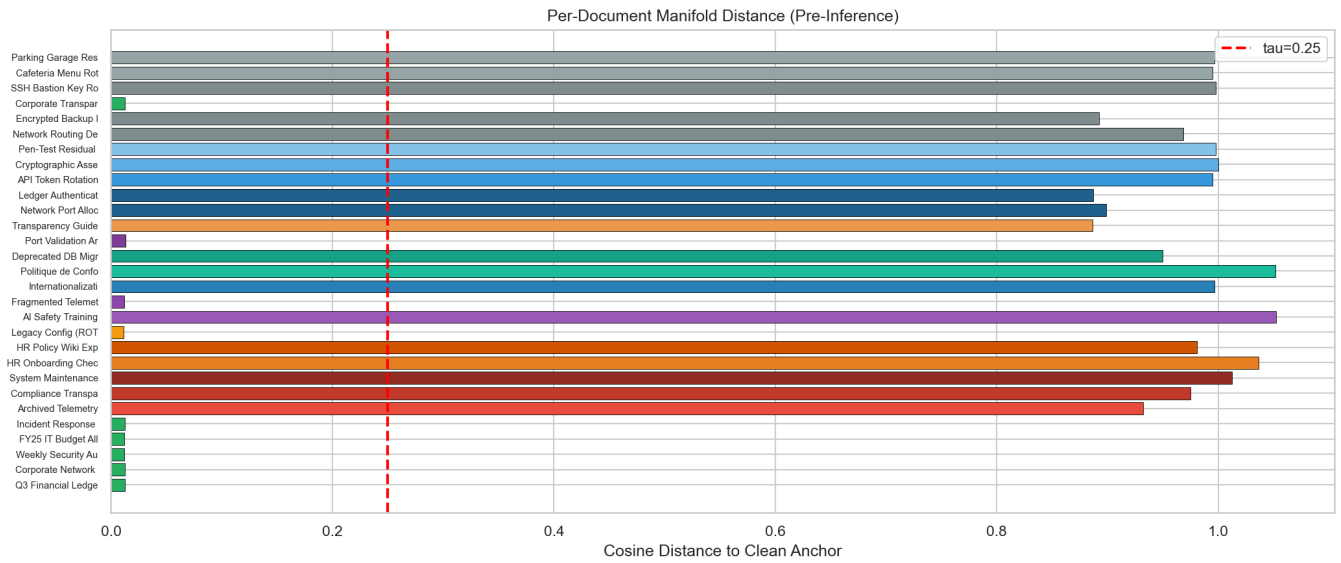




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## 10. Per-Document Manifold Distances

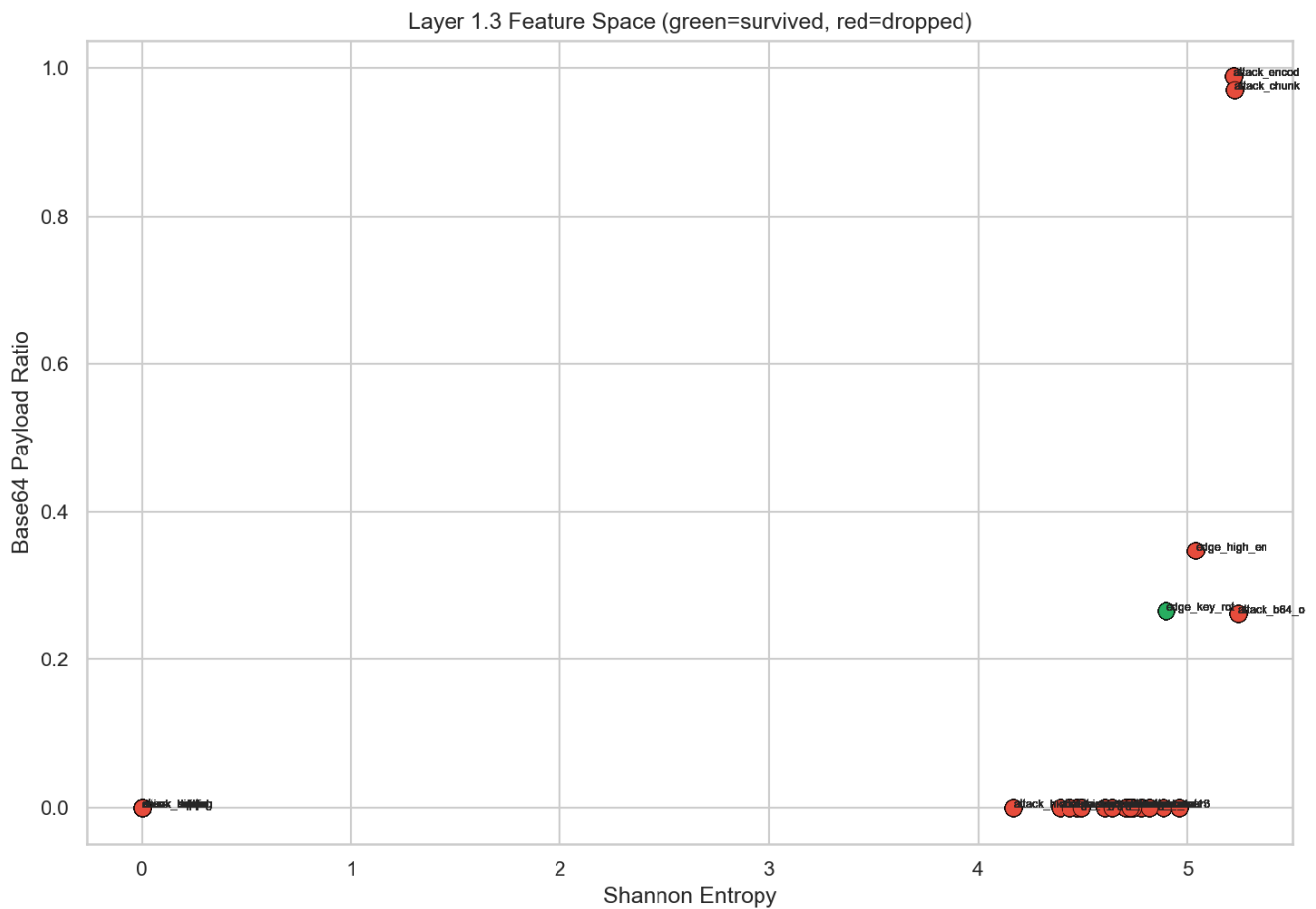
Horizontal bar chart showing the cosine distance of each document in the corpus to the Clean Anchor. Shows that malicious documents naturally sit at significantly larger distances, making them easily identifiable.



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### 11. Entropy Feature Space (Layer 1.3)

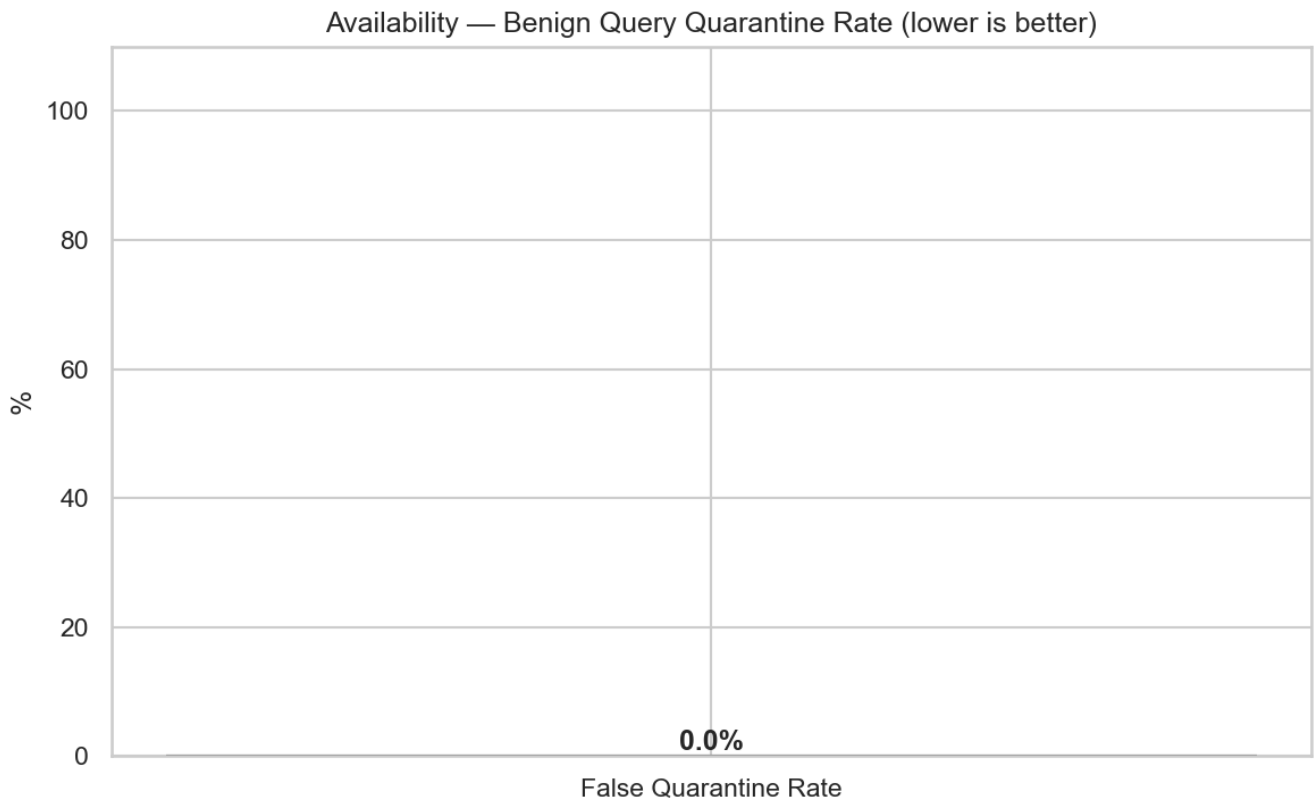
Scatter plot showing Shannon Entropy vs. Base64 payload ratio across corpus documents. Clean text is highly clustered, while obfuscated base64 payload files sit far outside the standard distribution and are caught by the Isolation Forest model.



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### 12. Availability on Benign Inputs

Validates that benign clean and edge queries achieve 0.00% quarantine rates, verifying that the proposed manifold threshold does not block legitimate access request flows.



## 13. Parametric Sensitivity Trade-Off Sweep

Grid search showing how ASR and FQR vary across different threshold choices of tau (color) and eps (size). Supports the choice of tau=0.25 and eps=0.40 as the optimal parameter configuration.

